



Reported outcome measures in complex fracture elbow dislocations: a systematic review

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Background: Complex elbow fracture dislocations, dislocation with fracture of one or several surrounding bony stabilizers, are difficult to manage and associated with poor outcomes. While many studies have explored treatment strategies but a lack of standardization of patient-reported outcome measures (PROMs) makes cross-study comparison difficult. In this systematic review, we aim to describe what injury patterns, measured outcomes, and associated complications are reported in the complex elbow fracture dislocation literature to provide outcome reporting recommendations that will facilitate improved future cross-study comparison.

Methods: A systematic review was performed per Preferred Reporting Items for Systematic reviews and Meta-Analyses guidelines. We queried PubMed/MEDLINE, Embase, Web of Science, and Cochrane databases to identify articles published between 2010 and 2022 reporting on adult patients who had a complex elbow fracture dislocation. Pathologic fractures were excluded. A bias assessment using the methodological index for nonrandomized studies criteria was conducted. For each article, patient demographics, injury pattern, outcome measures, and complications were recorded.

Results: Ninety-one studies reporting on 3664 elbows (3654 patients) with an elbow fracture and dislocation (weighted mean age 44 years, follow-up of 30 months, 41% female) were evaluated. Of these, the injury pattern was described in 3378 elbows and included 2951 (87%) terrible triad injuries and 72 (2%) transolecranon fracture-dislocations. The three most commonly reported classification systems were: Mason classification for radial head fractures, Regan and Morrey coronoid classification for coronoid fractures, and O'Driscoll classification for coronoid fractures. Range of motion was reported in 87 (96%) studies with most reporting flexion (n = 70), extension (n = 62), pronation (n = 68), or supination (n = 67). Strength was reported in 11 (12%) studies. PROMs were reported in 83 (91%) studies with an average of 2.6 outcomes per study. There were 14 outcome scores including the Mayo Elbow Performance Score (n = 69 [83%]), the Disabilities of Arm, Shoulder and Hand (DASH) score (n = 28 [34%]), the visual analog scale for pain (n = 27 [33%]), QuickDASH score (n = 13 [15.7%]), and Oxford Elbow score (n = 5 [6.0%]). No significance was found between the number of PROMs used per article and the year of publication ($P = .313$), study type ($P = .689$), complex fracture pattern ($P = .211$), or number of elbows included ($P = .152$).

Institutional Review Board approval was not required for this systematic review.

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Conclusion: There is great heterogeneity in reported PROMs in the complex elbow fracture dislocation literature. Although there is no gold standard PROM for assessing complex elbow fracture dislocations, we recommend the use of at least the Mayo Elbow Performance Score and DASH outcomes measures as well as visual analog scale pain rating scale in future studies to facilitate cross-study comparisons.

Level of evidence: Systematic Review; Research Methodology

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Keywords: Complex elbow; fracture dislocation; elbow; Monteggia; patient reported outcome measures; systematic review

The elbow is one of the most commonly dislocated joints of the upper extremity, with an incidence of six per 100,000 persons.¹¹⁹ Most dislocations are purely soft tissue injuries, however, roughly one-quarter of elbow dislocations include a concomitant elbow fracture.¹¹⁹ Complex elbow fracture dislocations are defined as dislocation with fracture of 1 or several bony stabilizers of the elbow joint, such as the radial head, proximal ulna, or coronoid process.¹⁰⁹ Surgical treatment of complex elbow fracture dislocations continues to be challenging and is at times associated with significant long-term morbidity and complications such as stiffness and instability.^{18,86,91} Treatment approach and outcomes vary by instability pattern, ranging from nonsurgical intervention to fracture fixation, joint replacement, and soft tissue repair in certain injuries.⁹⁸ This clinical challenge is compounded by a lack of consensus on how to report outcomes in this patient population.¹⁸ Understanding which outcomes are reported throughout the literature for complex elbow fracture dislocations is necessary to improve future approaches to management and ultimately patient care.

Inconsistencies in reporting clinical outcomes in complex elbow fracture dislocations make standardizing patient stratification and treatment decision-making challenging.³⁸ Many studies have explored the impact of different surgical and nonsurgical management approaches on clinical outcomes; however, cross-study comparisons are made difficult by these outcome measure inconsistencies.^{7,14,20,24} Others have investigated the correlation between patient-reported outcome measures (PROMs) and clinical outcome measurements, but to our knowledge, no comprehensive review of outcome measurements and clinical complications has been conducted for complex elbow fracture dislocations.^{13,51} Understanding this relationship will improve future research efforts and approaches to patient management through a pragmatic approach.

The purpose of this study was to conduct a comprehensive systematic review of articles related to complex elbow fracture dislocations and identify which outcomes are reported and the associated complications. We hypothesized that identified outcomes and complications would vary widely, independent of specific injury pattern. The goal for this data is to enable greater consistency of reported outcomes to permit comparability across future studies, ideally leading to improved management of

complex elbow fracture dislocations and enhanced patient outcomes.

Materials and methods

This systematic review was conducted under the guidelines set by the Preferred Reporting Items for Systematic Reviews and Meta-Analysis protocols criteria.¹¹⁰

Eligibility criteria

We included original studies reporting outcomes in patients with elbow fracture with concomitant dislocation. Studies were excluded for the following indications: being a duplicate, non-English text, only abstract available, review or meta-analysis, a case report or reported fewer than five patients, commentary or editorial, biomechanical or cadaveric study, or if outcomes were not reported (Fig. 1).^{1-3,5-8,11,12,14,16,17,19-25,27,28,32-35,39,41-43,45-47,50,52-55,58,60-68,71-79,81-85,87-89,92-96,99-103,105-108,111-115,117,118,120,121,123-126}

Search strategy

Cochrane Database of Systematic Reviews, Embase, PubMed, and Web of Science were searched in November 2022 for articles assessing complex elbow fracture — dislocations and post-operative outcomes published after January 2010. The search strategy included: ("terrible triad" OR "terrible-triad" OR "posterolateral rotatory") OR ("transolecranon" OR "trans-olecranon" OR "trans olecranon") OR ("Monteggia*") OR ("anteromedial" OR "antero medial" OR "antero-medial") AND "coronoid" AND "fracture") OR (("elbow" OR "ulna*" OR "radi*") AND ("fracture dislocation*" OR "fracture-dislocation*")). Articles were first screened by title and abstract for exclusion criteria by two authors (L.W., K.M.H.). If inclusion eligibility was questionable, the two authors included the study. Subsequent full text screening was conducted by the same authors. When there were any discrepancies or when a consensus could not be met, a panel of senior shoulder and elbow surgeons (J.J.K., B.S.S., T.W.W.) provided expert opinion.

Outcome measures and data extraction

Using a standardized data collection form, data extraction was performed by two authors (R.L.J., M.Q.B.). Characterizations of the articles extracted included the country of the study and whether the study was retrospective or prospective in nature.

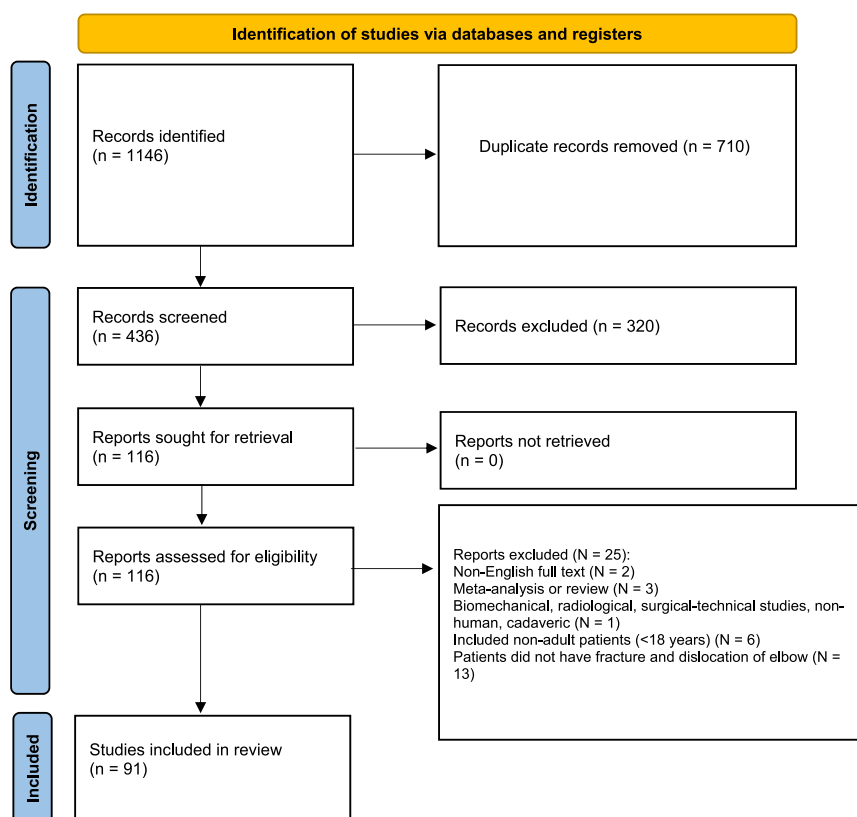


Figure 1 Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) flowchart of the study selection criteria.

General demographics of the included patients were collected and included age, sex, time to latest follow-up, and number of patients and elbows. Injury characteristics were extracted and included number of patients with predefined injury patterns (Monteggia, terrible triad, Monteggia-like lesions, transolecranon fracture-dislocation). Clinical outcomes that were included in the study were: arm strength, range of motion, functional outcomes, patient-reported or combined physician- and patient-reported outcome scores, and the incidence of complications.

Risk of bias assessment

The Methodological Index for Non-Randomized Studies (MINORS) criteria was utilized to assess bias by a single investigator (M.Q.B.). See [Supplementary Table](#) for MINORS scores.

Statistical analysis

Patient, article, fracture characteristics, and outcome measures were tabulated descriptively. We assessed the number of PROMs reported per article over time. Additionally, we assessed the number of PROMs reported per study based on year of publication, type of study (prospective vs. retrospective), complex fracture pattern, and number of elbows. We further assessed the usage of individual PROMs based on geography and year of publication. The geographic divisions comprised of North America, Europe, Asia, and other (encompassing South America, Africa, the Middle

East, and Oceania). The articles were grouped by publication year, utilizing intervals of 3 years. Temporal trends were evaluated with Cochran-Armitage tests and linear regression. Nontrend analysis involved the use of Fisher exact test and Welch's *t*-tests. Bonferroni post-hoc test was conducted for pairwise comparisons for significant interactions. All analyses were carried out using R software (version 4.2.0, R Core Team, Vienna, Austria). Significance was set at an α of 0.05.

Results

Search results

Our search strategy returned 1146 publications, of which 436 were found to be unique following duplicate exclusion ([Fig. 1](#)). We additionally excluded 320 articles during title and abstract screening, leaving 116 articles for full-text review. During full-text screening, 91 unique articles were found to meet inclusion criteria and included. We also completed a bias assessment using the MINORS criteria for retrospective studies.¹⁰⁴ The average MINORS score for comparative studies was 20.51 (out of 24) and for non-comparative studies was 11.78 (out of 16). See [Supplementary Table](#) for individual studies and MINORS scores.

Table 1 Characteristics of articles and included patients

Characteristic	Statistic	Studies reporting, % (N)
Article characteristics		
Top countries, % (N)		
China	25.3% (23)	100.0% (91)
United States	12.1% (11)	100.0% (91)
Top continents, % (N)		
Asia	39.6% (36)	100.0% (91)
Europe	28.6% (26)	100.0% (91)
North America	17.6% (16)	100.0% (91)
Study design, % (N)		
Randomized controlled trial	1.1% (1)	100.0% (91)
Prospective cohort	2.2% (2)	100.0% (91)
Retrospective cohort	2.2% (2)	100.0% (91)
Case-control	12.1% (11)	100.0% (91)
Case series	80.2% (73)	100.0% (91)
Patient demographics		
Mean elbows per study (median; range)	78.4 (24; 5 to 334)	100.0% (91)
Weighted mean age (range), yr	44 (34 to 75)	96.7% (88)
Weighted number of females per study (range)	17 (0 to 134)	95.6% (87)
Weighted % of females per study (range)	41% (0% to 70%)	95.6% (87)
Weighted follow-up length (range), mo	30 (6 to 135)	95.6% (87)
Weighted minimum follow-up (range), mo	17 (2 to 120)	96.7% (88)

Study characteristics

The senior author on most included studies was from China (23/91, 25%). The most common continents were Asia (36/91, 40%), Europe (26/91, 29%), and North America (16/91, 18%). The studies consisted of 1 (1.1%) randomized controlled trial, two (2.2%) prospective cohort studies, two (2.2%) prospective case series studies, two (2.2%) retrospective case-control studies, 11 (12%) retrospective cohort studies, and 73 (80%) retrospective case series studies (Table 1).

Patient characteristics

Overall, 91 articles were included, reporting on 3664 elbows (3654 patients). Weighted mean age was 44 years [range: 34-75 years], follow-up was 30 months [range: 6-135 months], and 41% were female [range: 0%-70%]

(Table 1). Of the reported elbow fracture-dislocations, 2951 (87.4%) were terrible triads, 257 (7.6%) were Monteggia-like patterns, 90 (2.7%) were Monteggia, 72 (2.1%) were transolecranon fracture-dislocation patterns. The remaining 286 elbows were not of predefined fracture-dislocation patterns and were a combination of fracture and dislocation of either the humerus, radius, or ulna at the elbow joint (Table 2).

Classifications

Eight different classification systems were reported in our data collection. Distal humeral fractures, radial fractures, and ulnar fractures were each individually classified with 1 classification system (AO foundation and Orthopaedic Trauma Association classification system, Mason Radial Head classification system, and AO foundation and Orthopaedic Trauma Association classification system, respectively). Coronoid fractures were classified using two systems, Regan and Morrey (64.8% of elbows, 59.3% of studies) and O'Driscoll classification systems (12.9% of elbows, 16.5% of studies). Monteggia fractures were classified in 22 studies utilizing three different systems including the Bado classification system (89.3% of elbows, 47.4% of studies), Jupiter classification system (83.6% of elbows, 57.9% of studies), and the Jupiter, Proximal Ulnar and Radial fracture-dislocation Comprehensive Classification Systems (38.9% of elbows, 10.5% of studies). Open fractures were classified by the Gustilo classification system (Table 2).

Range of motion

Surgeon measured range of motion (ROM) was reported in 87 (95.6%) studies. Of those, flexion ROM was reported in 70 (80.5%) studies, pronation ROM was reported in 68 (78.2%) studies, supination ROM was reported in 67 (77.0%) studies, extension ROM was reported in 62 (71.3%) studies, flexion-extension arc ROM was reported in 59 (67.8%) studies, pronation-supination arc ROM was reported in 46 (52.9%) studies, palmar flexion ROM was reported in two (2.3%) studies, and wrist extension ROM was reported in 1 (1.1%) study (Table 3).

Strength

Strength assessment was reported in 11 (12.1%) studies. Of those, grip strength was reported in six (54.5%) studies, manual muscle testing score was reported in two (18.2%) studies, and Biodex strength measurement was reported in 1 (9.1%) study (Table 3).

PROMs

PROMs were reported in 83 (91.2%) studies, with 2.6 PROMs per study on average (interquartile range, 2-3). Of

Table II Fracture characteristics and classifications

Characteristic	Elbows, % (N/Group total)	Studies, % (N)
Elbow fracture-dislocation of predefined pattern	92.2% (3378/3664)	94.5% (86/91)
Terrible triad	87.4% (2951/3378)	80.2% (69/86)
Monteggia	2.7% (90/3378)	10.5% (9/86)
Monteggia-like	7.6% (257/3378)	11.6% (10/86)
Transolecranon	2.1% (72/3378)	11.6% (10/86)
Non-predefined pattern of elbow fracture dislocation	7.8% (286/3664)	5.5% (5/91)
Humerus fracture classification		
AO/OTA classification	1.6% (59/3664)	3.3% (3/91)
Type A	20.3% (12/59)	66.7% (2/3)
Type B	20.3% (12/59)	100.0% (3/3)
Type C	59.3% (35/59)	100.0% (3/3)
Radius fracture classification		
Mason radial head classification	65.7% (2408/3664)	65.9% (60/91)
Mason Type I	17.4% (420/2408)	73.3% (44/60)
Mason Type II	37.7% (907/2408)	88.3% (53/60)
Mason Type III	36.6% (882/2408)	86.7% (52/60)
Mason Type IV	8.3% (199/2408)	15.0% (9/60)
Ulna fracture classification		
AO/OTA classification	0.3% (12/3664)	2.2% (2/91)
2U1	91.7% (11/12)	100.0% (2/2)
2U2	8.3% (1/12)	50.0% (1/2)
Coronoid fracture classification		
Regan and Morrey classification	64.8% (2373/3664)	59.3% (54/91)
Regan and Morrey Type I	44.2% (1050/2373)	90.7% (49/54)
Regan and Morrey Type II	40.7% (966/2373)	96.3% (52/54)
Regan and Morrey Type III	15.0% (357/2373)	68.5% (37/54)
O'Driscoll classification	12.9% (472/3664)	16.5% (15/91)
O'Driscoll type, not further specified	12.1% (57/472)	20.0% (3/15)
O'Driscoll Type I	38.6% (22/57)	66.7% (2/3)
O'Driscoll Type II	38.6% (22/57)	66.7% (2/3)
O'Driscoll Type III	22.8% (13/57)	66.7% (2/3)
Tip	50.2% (237/472)	66.7% (10/15)
Tip subtype 1	49.4% (117/237)	90.0% (9/10)
Tip subtype 2	50.6% (120/237)	80.0% (8/10)
Anteromedial	28.2% (133/472)	60.0% (9/15)
Anteromedial subtype 1	18.0% (24/133)	77.8% (7/9)
Anteromedial subtype 2	69.2% (92/133)	100.0% (9/9)
Anteromedial subtype 2	12.8% (17/133)	55.6% (5/9)
Basal	9.5% (45/472)	53.3% (8/15)
Basal subtype 1	13.3% (6/45)	50.0% (4/8)
Basal subtype 2	86.7% (39/45)	87.5% (7/8)
Monteggia fracture classification		
Bado classification	89.3% (310/347 [*])	47.4% (9/19 [*])
Bado classification type 1	15.8% (49/310)	77.8% (7/9)
Bado classification type 2	78.4% (243/310)	100.0% (9/9)
Bado classification type 3	3.9% (12/310)	55.6% (5/9)
Bado classification type 4	1.9% (6/310)	22.2% (2/9)
Jupiter classification	83.6% (290/347 [*])	57.9% (11/19 [*])
Jupiter classification type A	27.2% (79/290)	90.9% (10/11)
Jupiter classification type B	33.8% (98/290)	90.9% (10/11)
Jupiter classification type C	13.4% (39/290)	81.8% (9/11)
Jupiter classification type D	25.5% (74/290)	100.0% (11/11)
PURCCS classification	38.9% (135/347 [*])	10.5% (2/19 [*])
PURCCS type 1	4.4% (6/135)	50.0% (1/2)
PURCCS type 2	7.4% (10/135)	50.0% (1/2)
PURCCS type 3	3.0% (4/135)	50.0% (1/2)

(continued on next page)

Table II Fracture characteristics and classifications (*continued*)

Characteristic	Elbows, % (N/Group total)	Studies, % (N)
PURCCS type 4	4.4% (6/135)	50.0% (1/2)
PURCCS type 5	8.1% (11/135)	50.0% (1/2)
PURCCS type 6	1.5% (2/135)	50.0% (1/2)
PURCCS type A	4.4% (6/135)	50.0% (1/2)
PURCCS type B	33.3% (45/135)	100.0% (2/2)
PURCCS type C	27.4% (37/135)	100.0% (2/2)
PURCCS type D	1.5% (2/135)	50.0% (1/2)
PURCCS type E	4.4% (6/135)	100.0% (2/2)
Open fracture classification		
Gustilo classification	0.9% (34/3664)	8.8% (8/91)
Type I	55.9% (19/34)	62.5% (5/8)
Type II	14.7% (5/34)	50.0% (4/8)
Type III	14.7% (5/34)	12.5% (1/8)
Type IIIa	5.9% (2/34)	12.5% (1/8)
Type IIIb	5.9% (2/34)	12.5% (1/8)
Type IIIc	2.9% (1/34)	12.5% (1/8)

AO/OTA, AO foundation and Orthopaedic Trauma Association; PURCCS, Proximal Ulner and Radial fracture-dislocation Comprehensive Classification System.

* Monteggia and Monteggia-like studies aggregated.

those, the most common PROMs were the Mayo Elbow Performance Score (MEPS) ($n = 69$ [83.1%] studies), the Disabilities of Arm, Shoulder and Hand (DASH) score ($n = 28$ [33.7%] studies), the visual analog scale (VAS) ($n = 27$ [32.5%] studies), and the QuickDASH score ($n = 13$ [15.7%] studies), and Oxford Elbow score ($n = 5$ [6.0%] studies) (Table 3). The number of PROMs used per article did not increase significantly over time ($P = .294$; Fig. 2). There was no significant difference in the number of PROMs included based on year of publication ($P = .313$; Fig. 3A), study type ($P = .689$; Fig. 3B), complex fracture pattern ($P = .211$; Fig. 3C), and number of elbows ($P = .152$; Fig. 3D). The MEPS was the most commonly reported PROM in every continent except for North America (Fig. 4). The DASH score was the most commonly reported PROM in North America. The MEPS was the most commonly reported PROM in each 3-year increment (Fig. 5).

Physical exam reporting

Physical exam findings were reported in 35 (38.5%) studies. Of those, the most common were posteromedial rotary in varus instability test ($n = 25$ [52.1%] studies), valgus and varus instability test ($n = 15$ [31.3%] and $n = 12$ [25.0%] studies, respectively), and elbow lateral pivot-shift test $n = 9$.

Functional status

Functional outcomes were reported by 38 (41.8%) studies. Of those, satisfaction determination was reported in 26

(28.6%) of studies, time to return to activities of daily living was reported in 10 (11%) studies, time to return to work was reported in 12 (13%) studies, and time to return to activity or sport was reported in four (4%) studies (Table 3).

Complications

Complications were reported in all 91 studies (Table 4). The most common complications included heterotopic ossification ($n = 71$ [78%] studies), infection ($n = 53$ [58%] studies), instability ($n = 51$ [56%] studies), and ulnar nerve injury ($n = 50$ [55%] studies). Revision surgeries were reported by 66 (73%) studies (Table 4).

Discussion

Complex elbow fracture dislocations encompass a broad spectrum of injuries, ranging from terrible-triad injuries to various fracture-dislocation patterns, each posing a unique challenge in terms of management and recovery. As a result, the reported outcomes and complications following complex elbow trauma can vary significantly. Use of standardized outcome measures facilitate comparisons across studies and guide evidence-based treatment decisions. Traditionally, upper extremity outcome scores were developed for disease- or joint-specific assessment. Due to the scarcity of elbow-specific outcome measures, a diverse array of outcome assessments has been employed for elbow-related evaluations.^{40,122}

This review identified that PROMs were reported in 91% of included studies, with an average of 2.6 PROMs per

Table III Outcome measures reported by included studies

Outcome	% (N/Group total)
Patient-Reported Outcome Measures, % (N)	
Any PROM reported	91.2% (83/91)
MEPS	83.1% (69/83)
DASH score	33.7% (28/83)
Visual analog scale (VAS) for pain	32.5% (27/83)
QuickDASH	15.7% (13/83)
Oxford Elbow score	6.0% (5/83)
Mayo Modified Wrist score	3.6% (3/83)
ASES	3.6% (3/83)
Modified ASES	2.4% (2/83)
Morrey's Score	1.2% (1/83)
SF-36	1.2% (1/83)
Broberg and Morrey	1.2% (1/83)
Functional elbow score	1.2% (1/83)
Bruce score	1.2% (1/83)
Numeric pain score	1.2% (1/83)
Physical Exam Reporting, % (N)	
Any physical exam findings reported	52.7% (48/91)
Posteromedial rotatory in varus instability test	52.1% (25/48)
Excellent and good rating	39.6% (19/48)
Valgus test	31.3% (15/48)
Varus test	25.0% (12/48)
Pivot shift test	18.8% (9/48)
Hanging arm test	6.3% (3/48)
Chair raise test	4.2% (2/48)
Gravity test	4.2% (2/48)
Functional Status, % (N)	
Any functional outcomes reported	41.8% (38/91)
Satisfaction	68.4% (26/38)
Return to work	31.6% (12/38)
Return to activities of daily living	26.3% (10/38)
Return to activity or sports	10.5% (4/38)
Elbow range of motion, % (N)	
Any recorded	95.6% (87/91)
Flexion	80.5% (70/87)
Extension	71.3% (62/87)
Flexion-extension arc	67.8% (59/87)
Pronation	78.2% (68/87)
Supination	77.0% (67/87)
Pronation-supination arc	52.9% (46/87)
Flexion contracture	8.0% (7/87)
Palmar flexion	2.3% (2/87)
Wrist extension	1.1% (1/87)
Strength, % (N)	
Any recorded	12.1% (11/91)
Jamar Dynamometer (grip strength)	36.4% (4/11)
Manual muscle testing	18.2% (2/11)
Grip strength (method unspecified)	18.2% (2/11)
Biodex strength measurements (N·m)	9.1% (1/11)

PROM, Patient-Reported Outcome Measure; MEPS, Mayo Elbow Performance Score; DASH, Disabilities of the Arm, Shoulder and Hand; ASES, American Shoulder and Elbow Surgeons; SF-36, Short Form-36 Health Survey.

study. There were a total of 14 unique PROMs utilized across all studies, the most common of which were the MEPS (83%), DASH (34%), VAS (33%), QuickDASH (16%), and Oxford Elbow (6%) scores. A systematic review performed by Evans et al³⁸ also reported that the MEPS score is more commonly utilized than the DASH and QuickDASH scores in articles reporting any traumatic elbow injuries. The MEPS index offers the advantages of straightforward administration and the only required equipment is a goniometer. In contrast to the DASH and QuickDASH scores, which are patient-reported questionnaires employing a Likert scale, the MEPS score has a component of physician input potentially reducing bias. Though a large proportion of studies utilized an elbow specific score — MEPS, a substantial portion of the reported PROMs encompassed within this study consisted of outcome assessments primarily designed for the shoulder or regional scoring systems encompassing the entire upper extremity. This could be, in part, because there are only two frequently used elbow-specific outcome measures, the MEPS and the Oxford Elbow score.^{29,31} The MEPS was introduced by Broberg and Morrey in 1987,¹⁵ whereas the Oxford Elbow score was published in 2008.³¹ Both of these scores have undergone validation against other outcome measures^{9,29,56} and are adaptable for use in other languages.^{44,90,97} When assessing the two utilized elbow-specific outcome measures, the Oxford Elbow score is the only outcome measure that is solely patient reported and is thus a superior score to measure a PROM. However, given that we identified that the Oxford Elbow score was utilized substantially less when compared to the MEPS score (6% vs. 83%), this low utilization rate currently inhibits its use for adequate cross-study comparison.

Alternatively, the use of shoulder or regional scoring systems may indicate an intention to assess overall functional status of the entire limb. Conventionally, upper extremity outcome scores were developed to be disease or joint-specific but often correlate with disability in multiple parts of the upper extremity.³⁰ Surgeons often rely on the DASH and QuickDASH scores because they provide a comprehensive assessment of the function of the entire extremity, going beyond just the elbow. Regional scoring systems consider a broader range of upper limb functions, allowing surgeons to gain a more holistic understanding of postinjury recovery and the impact on overall arm and hand function. Given the lack of a gold standard outcome measure for complex elbow fracture dislocations, the authors underscore the importance of future studies incorporating at least one elbow and upper extremity-specific PROM instead of solely relying on quality of life or pain scores to gain a more comprehensive understanding of overall patient function and to allow better cross-study comparison.

Given the findings of this study, the authors recommend the use of at least three outcome measures, preferably the DASH score, MEPS index, and the VAS score for pain rating, in future studies to facilitate cross-study

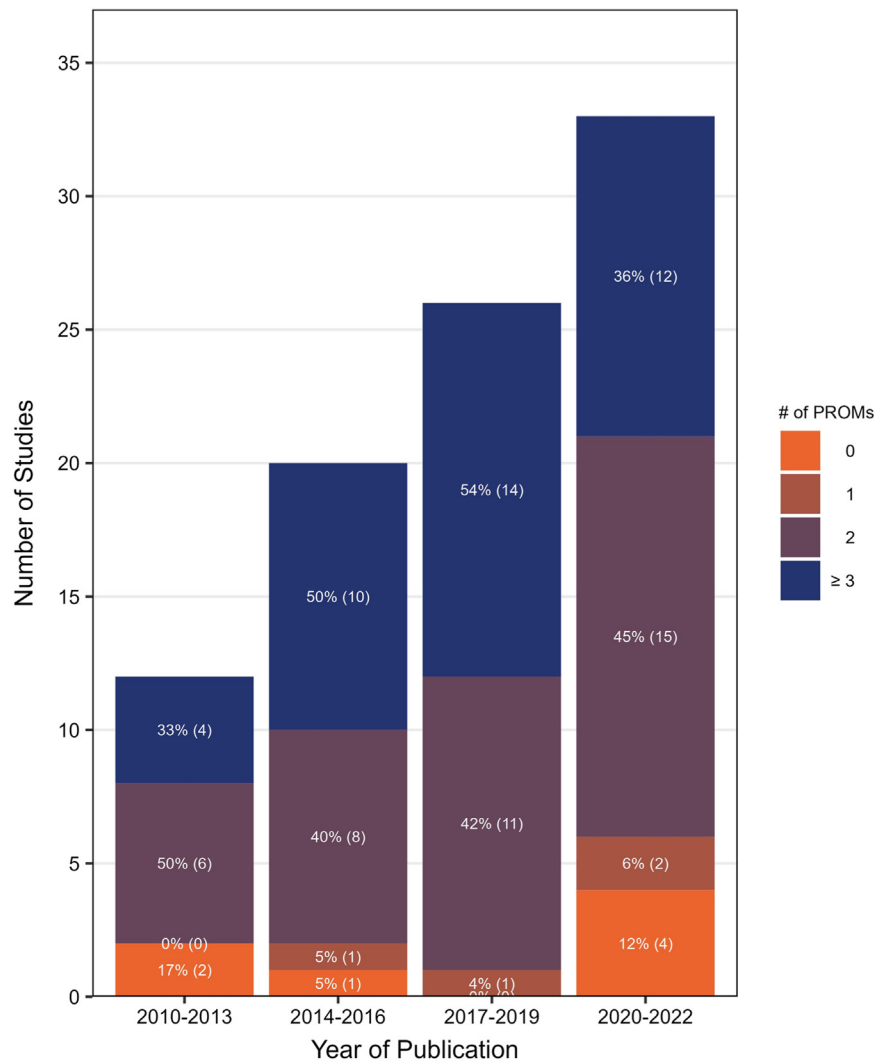


Figure 2 Trends in number of patient-reported outcomes (PROMs) included in each study over time stratified by 3-year increments.

comparisons and allow for better evidence-based decision making. The MEPS is the most commonly used elbow-specific outcome score and has been validated to reflect patient satisfaction.^{31,37,80} The DASH score has been shown to not only correlate with pain levels⁵⁹ but is also valid and reliable when compared with joint-specific outcome measures.¹⁰ Moreover, the DASH score has been validated to measure outcomes following elbow injury³¹ and correlate with patient satisfaction following simple elbow dislocations.⁴ If there is concern for questionnaire burden, the QuickDASH may be used instead of the DASH score.⁴⁸ While susceptible to potential influences from psychosocial factors,⁵⁷ the VAS pain assessment remains a straightforward single-item tool, facilitating simple administration, the tracking of pain fluctuations over time, and interpatient comparisons.

We observed that 96% of included studies documented reported ROM. This is intuitive, as stiffness is one of the most common complications following elbow trauma⁶⁹ and a greater deficit in flexion-extension arc is correlated with lower DASH, Oxford Elbow, and patient satisfaction scores.⁴ Anakwe et al⁴ noted the importance of reporting an overall arc of motion, as patients tend to tolerate loss of terminal elbow extension better than the loss of flexion. Only 12% of included studies reported measurements of strength, of which grip strength was the most prevalent ($n = 6$), followed by manual muscle testing ($n = 2$). Grip strength is primarily attributed to the activation of wrist and finger flexor muscles, with some of these muscles originating at the elbow. Consequently, in cases of complex elbow fracture dislocations, these crucial muscles may be at risk of compromise. Higher patient satisfaction scores have been correlated with

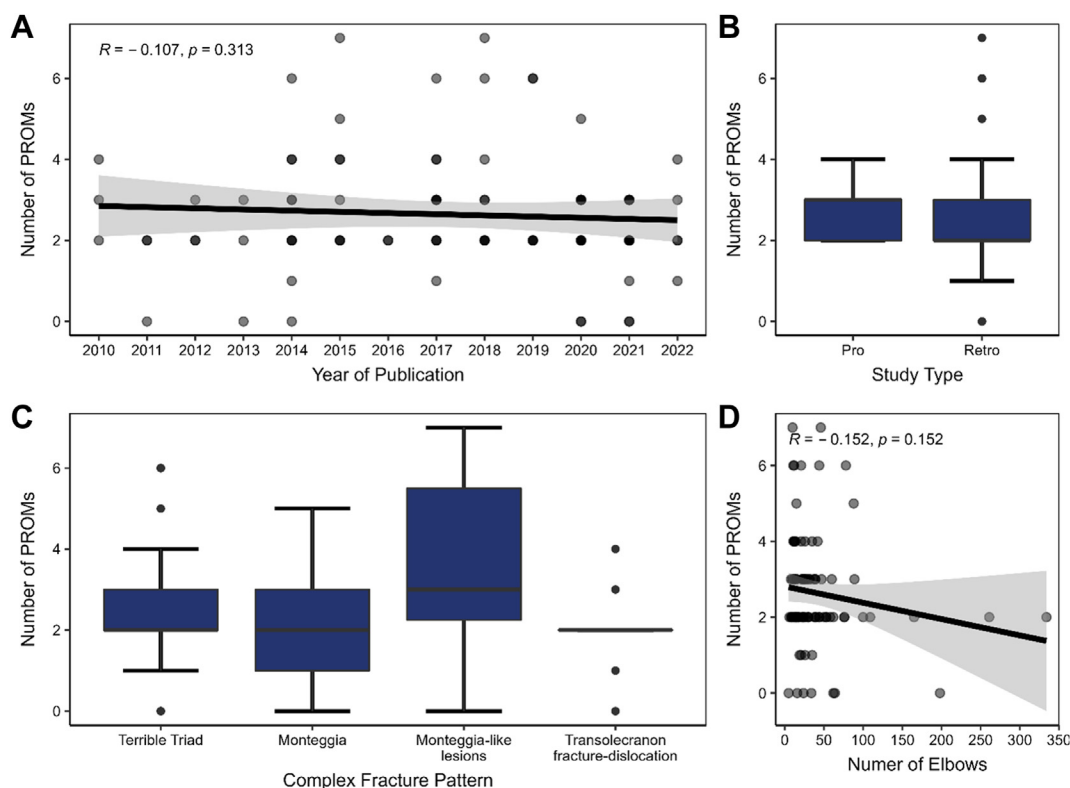


Figure 3 Trends in number of patient-reported outcome measures (PROMs) based on year of publication (A), study type (B), complex fracture pattern (C), and number of elbows (D).

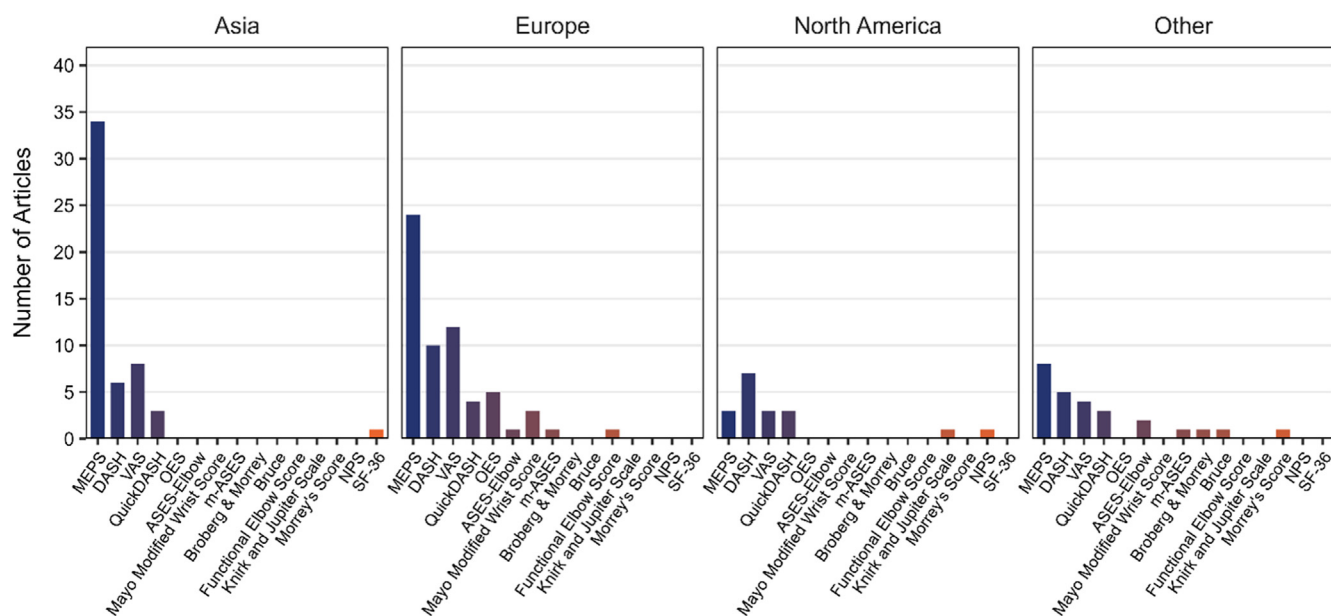


Figure 4 Trends in patient-reported outcome (PROM) use, stratified by geographic region of corresponding author.

improved grip strength²⁶ and a more favorable elbow flexion-extension arc⁴⁹. The authors of this review advocate reporting grip strength measurements alongside assessments

of the flexion-extension and pronation-supination arcs in studies evaluating the complex elbow, as favorable outcomes in these assessments have close association with improved

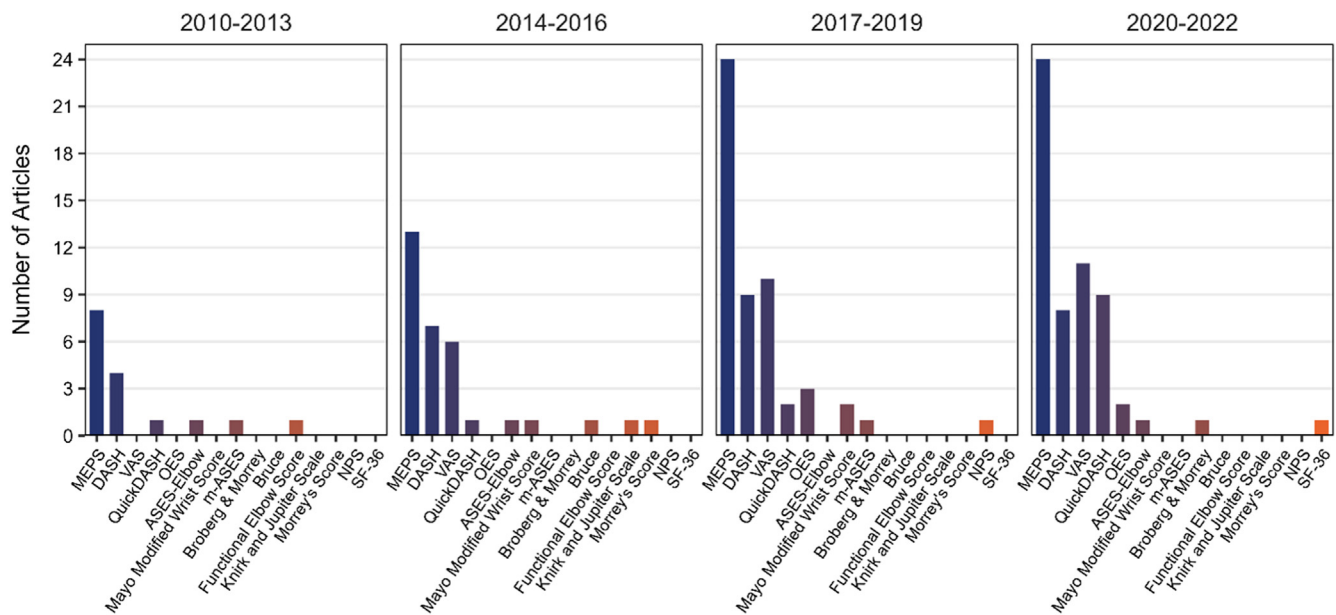


Figure 5 Trends in reporting of specific patient-reported outcomes (PROMs) over time, stratified by 3-year increments.

Table IV Complications and revisions reported by included studies

Complications, % (N)	
Mention of complications	100.0% (91/91)
Heterotopic ossification	78.0% (71/91)
Infection	58.2% (53/91)
Instability	56.0% (51/91)
Ulnar nerve injury	54.9% (50/91)
Stiffness	42.9% (39/91)
Post-traumatic arthritis	41.8% (38/91)
Malunion	41.8% (38/91)
Hardware failure	29.7% (27/91)
Dislocation	25.3% (23/91)
Loosening	23.1% (21/91)
Increased pain	15.4% (14/91)
Degenerative changes	13.2% (12/91)
Radial nerve injury	13.2% (12/91)
Post-traumatic synostosis	12.1% (11/91)
Nerve injury (unspecified)	9.9% (9/91)
Overstuffing	9.9% (9/91)
Healing complication	7.7% (7/91)
Complex regional pain syndrome	6.6% (6/91)
Osteolysis	6.6% (6/91)
Ectopic ossification	5.5% (5/91)
Re-fracture	4.4% (4/91)
Median nerve injury	4.4% (4/91)
Distal humerus necrosis	4.4% (4/91)
Compartment syndrome	3.3% (3/91)
Loss of strength	1.1% (1/91)
Radial head necrosis	1.1% (1/91)
Revision	72.5% (66/91)

DASH, Disabilities of the Arm, Shoulder and Hand; ASES, American Shoulder and Elbow Surgeons; SF-36, Short Form-36 Health Survey.

ability to perform activities of daily living⁷⁰ and prevention of long term deficit following elbow dislocation.^{4,36,116}

Several studies reported instability, of which the most common sources of instability were varus posteromedial instability (n = 25), followed by valgus instability (n = 15) and varus instability (n = 12). Persistent elbow instability is a known complication following complex elbow fracture dislocations, typically resulting from disruption to the medial and lateral ligamentous complex causing varus or valgus instability, respectively.⁶⁹ Functional outcomes, including time to return to activities of daily living, return to work, and return to activity/sport, were reported in 26% of studies. The authors advocate the inclusion of functional status, as patients often inquire when they can expect to resume their normal activities following an injury. By reporting the time it takes to return to work, perform daily tasks, and participate in sport or exercise, researchers and clinicians can compare results and better elucidate which interventions will hasten return to their baseline level of activity.

This study is the first systematic review article to our knowledge examining outcomes following complex elbow fracture dislocations. However, it is important to consider its inherent limitations. Despite utilizing highly queried database with broad search terms, articles may have been missed. Further, compounding bias and selection bias are concerns with review articles, and the analysis of many retrospective studies dictates the overall conclusions that can be made. We did not set a minimum level of evidence for inclusion. The primary objective of this study was to conduct a comprehensive review of reported outcomes for complex elbow fracture dislocations, and consequently,

this did not include evaluation of the validity, accuracy, or clinical relevance of the included PROMs. We chose to include studies over a recent 12-year period to accurately reflect the latest outcome measures and changes in practice.

Conclusion

In summary, our systematic review demonstrated significant heterogeneity in PROMs reported in complex elbow fracture dislocation literature currently preventing cross-study comparison between interventions and outcomes. While there is no definitive gold standard PROM for assessing complex elbow fracture dislocations, we advocate for at least the inclusion of the MEPS index, the DASH score, the VAS pain rating scale, functional status, grip strength, and surgeon measured ROM in future studies to facilitate cross-study comparison. Complex elbow fracture dislocations represent a broad spectrum of elbow pathology; future standardization of PROMs will allow clinicians to better assess outcomes following both operative and nonoperative management.

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Supplementary Data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jse.2024.02.038>

References

1. Afifi A, Lymona AM, Galal S. Radial head fixation vs replacement in terrible triad: preliminary results of a prospective cohort study with patient reported outcome. *Indian J Orthop* 2020;54:254-9. <https://doi.org/10.1007/s43465-020-00090-x>
2. AlQahtani S, Aibinder WR, Parry JA, Seltser A, Athwal GS, King GJW, et al. Static and dynamic external fixation are equally effective for unstable elbow fracture-dislocations. *J Orthop Trauma* 2021;35:e82-8. <https://doi.org/10.1097/BOT.0000000000001876>
3. Álvarez Muñoz M, Pardo García JM, García Lamas L, Porras Moreno M, Jiménez Díaz V, Cecilia López D. Protocolised surgical treatment of terrible triad of elbow. Results and complications. *Rev Esp Cir Ortop Traumatol Engl Ed* 2019;63:281-8. <https://doi.org/10.1016/j.recot.2019.02.003>
4. Anakwe RE, Middleton SD, Jenkins PJ, McQueen MM, Court-Brown CM. Patient-reported outcomes after simple dislocation of the elbow. *J Bone Joint Surg Am* 2011;93:1220-6. <https://doi.org/10.2106/JBJS.J.00860>
5. Antoni M, Eichler D, Kempf J-F, Clavert P. Anterior capsule re-attachment in terrible triad elbow injury with coronoid tip fracture. *Orthop Traumatol Surg Res* 2019;105:1575-83. <https://doi.org/10.1016/j.otsr.2019.09.024>
6. Antoni M, Kempf J-F, Clavert P. Comparison of bipolar and monopolar radial head prostheses in elbow fracture-dislocation. *Orthop Traumatol Surg Res* 2020;106:311-7. <https://doi.org/10.1016/j.otsr.2019.10.027>
7. Antoni M, Mereb T, Ginot G, Meyer N, Clavert P. Prognostic factors for traumatic elbow osteoarthritis after terrible triad surgery, and functional impact. *Orthop Traumatol Surg Res* 2021;107:102826. <https://doi.org/10.1016/j.otsr.2021.102826>
8. de Aquino Santos A, Tonelli TA, Matsunaga FT, Matsumoto MH, Netto NA, Tamaoki MJS. Result from surgical treatment on the terrible triad of the elbow. *Rev Bras Ortop* 2015;50:403-8. <https://doi.org/10.1016/j.rboe.2015.06.010>
9. Baessler A, Eason RR, Joyce MR, Dibaba DT, Wan JY, Azar FM, et al. Reliability testing of the Mayo Elbow Performance Score in post-operative patients. *J Surg Orthop Adv* 2022;31:229-32.
10. Beaton DE, Katz JN, Fossel AH, Wright JG, Tarasuk V, Bombardier C. Measuring the whole or the parts? Validity, reliability, and responsiveness of the disabilities of the arm, shoulder and hand outcome measure in different regions of the upper extremity. *J Hand Ther* 2001;14:128-46.
11. Beingsnessner DM, Nork SE, Agel J, Viskontas D. A fragment-specific approach to type IID Monteggia elbow fracture-dislocations. *J Orthop Trauma* 2011;25:414-9. <https://doi.org/10.1097/BOT.0b013e3181fc6255>
12. Ben Abdellah A, Ben Salah S, Darraz S, Tebbaa A, Jelti O, Mokhtari O, et al. Terrible triad injury of the elbow: a PROCESS-compliant surgical case series from Eastern Morocco. *Ann Med Surg* 2022;78:103914. <https://doi.org/10.1016/j.amsu.2022.103914>
13. Bhashyam AR, Ochen Y, Van Der Vliet QMJ, Leenen LPH, Hietbrink F, Houwert RM, et al. Association of patient-reported outcomes with clinical outcomes after distal humerus fracture treatment. *J Am Acad Orthop Surg Glob Res Rev* 2020;4:e19.00122. <https://doi.org/10.5435/JAAOSGlobal-D-19-00122>
14. Brigato RM, Mouraria GG, Kikuta FK, Coelho SDP, Cruz MA, Zoppi Filho A. Functional evaluation of patients with surgically treated terrible triad of the elbow. *Acta Ortopédica Bras* 2015;23:138-41. <https://doi.org/10.1590/1413-78522015230301008>
15. Broberg MA, Morrey BF. Results of treatment of fracture-dislocations of the elbow. *Clin Orthop* 1987;216:109-19.
16. Calderazzi F, Menozzi M, Nosenzo A, Galavotti C, Solinas S, Vaienti E, et al. Monteggia-like lesions: preliminary reports and mid-term results of a single center. *Acta Biomed* 2020;91:e2020134. <https://doi.org/10.23750/abm.v91i4.9864>
17. Chan K, MacDermid JC, Faber KJ, King GJW, Athwal GS. Can we treat select terrible triad injuries nonoperatively? *Clin Orthop* 2014;472:2092-9. <https://doi.org/10.1007/s11999-014-3518-9>
18. Chan K, King GJW, Faber KJ. Treatment of complex elbow fracture-dislocations. *Curr Rev Musculoskelet Med* 2016;9:185-9. <https://doi.org/10.1007/s12178-016-9337-8>
19. Chemama B, Bonneville N, Peter O, Mansat P, Bonneville P. Terrible triad injury of the elbow: how to improve outcomes? *Orthop Traumatol Surg Res* 2010;96:147-54. <https://doi.org/10.1016/j.rcot.2010.02.008>
20. Chen H-W, Bi Q. Surgical outcomes and complications in treatment of terrible triad of the elbow: comparisons of 3 surgical approaches.

- Med Sci Monit 2016;22:4354-62. <https://doi.org/10.12659/msm.897297>
21. Chen H-W, Huang S-M. Risk factors of efficacy for patients receiving surgical treatment following terrible triad of the elbow joint: a comparative study. *Medicine (Baltim)* 2019;98:e13836. <https://doi.org/10.1097/MD.00000000000013836>
 22. Chen H-W, Liu G, Ou S, Fei J, Zhao G, Wu L-J, et al. Operative treatment of terrible triad of the elbow via posterolateral and anteromedial approaches. *PLoS One* 2015;10:e0124821. <https://doi.org/10.1371/journal.pone.0124821>
 23. Chen H-W, Wang Z-Y, Wei G-S. Evaluation of the therapeutic effects of lateral approach combined with anteromedial approach in the treatment of terrible triad of the elbow. *Biomed Pharmacother* 2017; 91:748-54. <https://doi.org/10.1016/j.biopha.2017.04.007>
 24. Chen AC-Y, Cheng Y-H, Chiu C-H, Cheng C-Y, Chan Y-S. Long-term outcomes of radial head arthroplasty in complex elbow fracture dislocation. *J Clin Med* 2021;10:3488. <https://doi.org/10.3390/jcm10163488>
 25. Cho E, Weber MB, Opel D, Lee A, Hoyer H, Bafus BT. Complications and functional outcomes after transolecranon distal humerus fracture. *J Shoulder Elbow Surg* 2021;30:479-86. <https://doi.org/10.1016/j.jse.2020.07.012>
 26. Chung KC, Haas A. Relationship between patient satisfaction and objective functional outcome after surgical treatment for distal radius fractures. *J Hand Ther* 2009;22:302-8. <https://doi.org/10.1016/j.jht.2009.04.007>
 27. Corbet C, Boudissa M, Dao Lena S, Ruatti S, Corcella D, Tonetti J. Surgical treatment of terrible triad of the elbow: retrospective continuous 50-patient series at 2 years' follow-up. *Orthop Traumatol Surg Res* 2023;109:103057. <https://doi.org/10.1016/j.otsr.2021.103057>
 28. Cui H-M, Yu Y-L, He Y, Cheng Y, Liu J-Z, Zheng W, et al. Management of elbow stiffness after postoperative treatment of terrible triad elbow injury: maintaining mobility and stability using a combined protocol. *Int Orthop* 2018;42:609-18. <https://doi.org/10.1007/s00264-017-3721-x>
 29. Cusick MC, Bonnaig NS, Azar FM, Mauck BM, Smith RA, Throckmorton TW. Accuracy and reliability of the Mayo Elbow Performance Score. *J Hand Surg* 2014;39:1146-50. <https://doi.org/10.1016/j.jhsa.2014.01.041>
 30. Davis AM, Beaton DE, Hudak P, Amadio P, Bombardier C, Cole D, et al. Measuring disability of the upper extremity: a rationale supporting the use of a regional outcome measure. *J Hand Ther* 1999;12: 269-74.
 31. Dawson J, Doll H, Boller I, Fitzpatrick R, Little C, Rees J, et al. The development and validation of a patient-reported questionnaire to assess outcomes of elbow surgery. *J Bone Joint Surg Br* 2008;90-B: 466-73. <https://doi.org/10.1302/0301-620x.90b4.20290>
 32. Desai MJ, Matson AP, Ruch DS, Leversedge FJ, Aldridge JM, Richard MJ. Perioperative Glucocorticoid administration improves elbow motion in terrible triad injuries. *J Hand Surg* 2017;42:41-6. <https://doi.org/10.1016/j.jhsa.2016.11.011>
 33. Doms P, Griffiths E, White A. Outcomes following surgical management of complex terrible triad injuries of the elbow: a single surgeon case series. *Shoulder Elb* 2018;10:216-22. <https://doi.org/10.1177/1758573217713694>
 34. Du S, Wei L, He B, Fang Z, Zhou E, Ma X, et al. Dynamic fixation using rigid tape in rehabilitation after surgery of terrible triad injury of the elbow: a randomized trial. *J Back Musculoskelet Rehabil* 2021;34:957-64. <https://doi.org/10.3233/BMR-200279>
 35. Eden L, Frey SP, Gilbert F, Jordan MC, Fenwick A, Meffert RH. Anatomically shaped locking plates for radial head and olecranon fracture fixation in Monteggia-like lesions. *Technol Health Care* 2020;28:193-201. <https://doi.org/10.3233/THC-191812>
 36. Egol KA, Immerman I, Paksima N, Tejwani N, Koval KJ. Fracture-dislocation of the elbow functional outcome following treatment with a standardized protocol. *Bull NYU Hosp Jt Dis* 2007;65:263-70.
 37. Engelke J, Vorm Walde M, Schnetzke M, Grützner P-A, Nolte P-C. High correlation of the subjective elbow value with Mayo Elbow Performance Score and Oxford Elbow score in patients with elbow dislocation. *JSES Int* 2023;7:868-71. <https://doi.org/10.1016/j.jseint.2023.04.010>
 38. Evans JP, Smith CD, Fine NF, Porter I, Gangannagaripalli J, Goodwin VA, et al. Clinical rating systems in elbow research-a systematic review exploring trends and distributions of use. *J Shoulder Elbow Surg* 2018;27:e98-106. <https://doi.org/10.1016/j.jse.2017.12.027>
 39. Fitzgibbons PG, Louie D, Dyer GSM, Blazar P, Earp B. Functional outcomes after fixation of "terrible triad" elbow fracture dislocations. *Orthopedics* 2014;37:e373-6. <https://doi.org/10.3928/01477447-2014-0401-59>
 40. Flynn L, Mamelson K, Schoch BS, Wright TW, King JJ. Outcome measures utilized in the Capitellum and Trochlea fracture literature: a systematic review. *J Hand Surg Glob Online* 2019;1:144-8. <https://doi.org/10.1016/j.jhsg.2019.05.002>
 41. Galbiatti JA, Cardoso FL, Ferro JAS, Godoy RCG, Belluci S de OB, Palacio EP. Terrible triad of the elbow: evaluation of surgical treatment. *Rev Bras Ortop* 2018;53:460-6. <https://doi.org/10.1016/j.rboe.2018.05.012>
 42. Garrigues GE, Wray WH, Lindenhovius ALC, Ring DC, Ruch DS. Fixation of the coronoid process in elbow fracture-dislocations. *J Bone Joint Surg Am* 2011;93:1873-81. <https://doi.org/10.2106/JBJS.1.01673>
 43. Gereli A, Nalbantoğlu U, Dikmen G, Seyhan M, Türkmen M. Fracture-dislocations of the proximal ulna. *Acta Orthop Traumatol Turc* 2015;49:233-40. <https://doi.org/10.3944/AOTT.2015.14.0178>
 44. Giannaka M, Andriagannakis A, Dimitriadis Z, Fandridis E, Kapreli E, Strimpakos N. Cross-cultural validation of the Oxford Elbow score and Mayo Elbow Performance Score in Greek. *Musculoskelet Sci Pract* 2022;57:102499. <https://doi.org/10.1016/j.msksp.2021.102499>
 45. Giannicola G, Scacchi M, Sacchetti FM, Cinotti G. Clinical usefulness of proximal ulnar and radial fracture-dislocation comprehensive classification system (PURCCS): prospective study of 39 cases. *J Shoulder Elbow Surg* 2013;22:1729-36. <https://doi.org/10.1016/j.jse.2013.07.050>
 46. Giannicola G, Calella P, Piccioli A, Scacchi M, Gumina S. Terrible triad of the elbow: is it still a troublesome injury? *Injury* 2015;46: S68-76. [https://doi.org/10.1016/S0020-1383\(15\)30058-9](https://doi.org/10.1016/S0020-1383(15)30058-9)
 47. Gonçalves LBJ, Neto J de AES, Correa Filho MRC, de Andrade RP, de Andrade MAP, Gomes AH, et al. Terrible triad of the elbow: influence of radial head treatment. *Rev Bras Ortop* 2014;49:328-33. <https://doi.org/10.1016/j.rboe.2014.07.001>
 48. Gummesson C, Ward MM, Atroshi I. The shortened disabilities of the arm, shoulder and hand questionnaire (QuickDASH): validity and reliability based on responses within the full-length DASH. *BMC Musculoskelet Disord* 2006;7:44. <https://doi.org/10.1186/1471-2474-7-44>
 49. Gundes H, Selekt Ö, Gök U, Gumuslu B, Buluc L. The relation between elbow range of motion and patient satisfaction after open release of stiff elbow. *Acta Orthop Traumatol Turc* 2017;51:303-7. <https://doi.org/10.1016/j.aott.2017.05.005>
 50. Gupta A, Barei D, Khwaja A, Beingsessner D. Single-staged treatment using a standardized protocol results in functional motion in the majority of patients with a terrible triad elbow injury. *Clin Orthop* 2014;472:2075-83. <https://doi.org/10.1007/s11999-014-3475-3>
 51. de Haan J, Schep N, Tuinebreijer W, den Hartog D. Complex and Unstable simple elbow dislocations: a review and Quantitative analysis of individual patient data. *Open Orthop J* 2010;4:80-6. <https://doi.org/10.2174/1874325001004020080>
 52. Haller JM, Hulet DA, Hannay W, Cardon J, Tashjian R, Beingsessner D. Patient outcomes after transolecranon fracture-dislocation. *J Am Acad Orthop Surg* 2021;29:109-15. <https://doi.org/10.5435/JAAOS-D-20-00254>
 53. He X, Fen Q, Yang J, Lei Y, Heng L, Zhang K. Risk factors of elbow stiffness after open Reduction and internal fixation of the terrible triad of the elbow joint. *Orthop Surg* 2021;13:530-6. <https://doi.org/10.1111/os.12879>

54. Hou F, Liang X, Fan W, Zhao F, Li D. Analysis of twenty-five cases of terrible triad injury of the elbow surgically treated with a single lateral approach. *Int Orthop* 2021;45:241-6. <https://doi.org/10.1007/s00264-020-04891-5>
55. Ikemoto RY, Murachovsky J, Bueno RS, Nascimento LGP, Carmargo AB, Corrêa VE. Terrible triad of the elbow: functional results of surgical treatment. *Acta Ortop. Bras* 2017;25:283-6. <https://doi.org/10.1590/1413-785220172506168821>
56. Iordens GIT, Den Hartog D, Tuinebreijer WE, Eygendaal D, Schep NWL, Verhofstad MHJ, et al. Minimal important change and other measurement properties of the Oxford Elbow Score and the Quick Disabilities of the Arm, Shoulder, and Hand in patients with a simple elbow dislocation: validation study alongside the multicenter FuncSiE trial. *PLoS One* 2017; 12:e0182557. <https://doi.org/10.1371/journal.pone.0182557>
57. Jamison RN, Gracely RH, Raymond SA, Levine JG, Marino B, Herrmann TJ, et al. Comparative study of electronic vs. paper VAS ratings: a randomized, crossover trial using healthy volunteers. *Pain* 2002;99:341-7. [https://doi.org/10.1016/s0304-3959\(02\)00178-1](https://doi.org/10.1016/s0304-3959(02)00178-1)
58. Jeong W-K, Oh J-K, Hwang J-H, Hwang S-M, Lee W-S. Results of terrible triads in the elbow: the advantage of primary restoration of medial structure. *J Orthop Sci* 2010;15:612-9. <https://doi.org/10.1007/s00776-010-1515-8>
59. Jester A, Harth A, Germann G. Measuring levels of upper-extremity disability in employed adults using the DASH questionnaire. *J Hand Surg* 2005;30:1074.e1-e10. <https://doi.org/10.1016/j.jhsa.2005.04.009>
60. Jiménez-Martín A, Contreras-Joya M, Navarro-Martínez S, Najarro-Cid FJ, Santos-Yubero FJ, Pérez-Hidalgo S. Clinical results of radial arthroplasty in Hotchkiss' terrible triad, a case series of 47. *Rev Esp Cir Ortop Traumatol Engl Ed* 2020;64:83-91. <https://doi.org/10.1016/j.recot.2019.11.003>
61. Jung S-W, Kim D-H, Kang S-H, Eho Y-J, Yang S-W, Lee G-E. Risk factors that influence Subsequent Recurrent instability in terrible triad injury of the elbow. *J Orthop Trauma* 2019;33:250-5. <https://doi.org/10.1097/BOT.0000000000001425>
62. Jung M, Groetzner-Schmidt C, Porschke F, Grützner PA, Guehring T, Schnetzke M. Monteggia-like lesions in adults treated with radial head arthroplasty-mid-term follow-up of 27 cases. *J Orthop Surg* 2020;15:5. <https://doi.org/10.1186/s13018-019-1540-z>
63. Jungbluth P, Tanner S, Schnependahl J, Grassmann J-P, Wild M, Hakimi M, et al. The challenge of Monteggia-like lesions of the elbow: mid-term results of 46 cases. *Bone Jt J* 2018;100-B:212-8. <https://doi.org/10.1302/0301-620X.100B2.BJJ-2017-0398.R2>
64. Kaneshiro Y, Yano K, Yokoi T, Sakanaka H, Fukuda M, Hidaka N. Is repair of a small coronoid fracture required in the surgical treatment of terrible triad injury of the elbow? *J Hand Surg Asian Pac Vol* 2022;27: 345-51. <https://doi.org/10.1142/S2424835522500412>
65. Kim B-S, Kim D-H, Byun S-H, Cho C-H. Does the coronoid always need to be fixed in terrible triad injuries of the elbow? Mid-term postoperative outcomes following a standardized protocol. *J Clin Med* 2020;9:3500. <https://doi.org/10.3390/jcm9113500>
66. Klug A, Konrad F, Gramlich Y, Hoffmann R, Schmidt-Horlohé K. Surgical treatment of the radial head is critical to the outcome of Monteggia-like lesions. *Bone Joint J* 2019;101-B:1512-9. <https://doi.org/10.1302/0301-620X.101B12.BJJ-2019-0547.R1>
67. Klug A, Nagy A, Gramlich Y, Hoffmann R. Surgical treatment of the radial head is crucial for the outcome in terrible triad injuries of the elbow. *Bone Joint J* 2020;102-B:1620-8. <https://doi.org/10.1302/0301-620X.102B12.BJJ-2020-0762.R1>
68. Laun R, Wild M, Brosius L, Hakimi M. Monteggia-like lesions - treatment strategies and one-year results. *GMS Interdiscip. GMS Interdiscip Plast Reconstr Surg DGPW* 2015;4:Doc13. <https://doi.org/10.3205/ips000072>
69. Layson J, Best BJ. Elbow dislocation. In: StatPearls. Treasure Island (FL). StatPearls Publishing; 2023. Available from: <http://www.ncbi.nlm.nih.gov/books/NBK549817/>
70. Lee KW, Kim YI, Kim HY, Yang DS, Lee GS, Choy WS. Three-dimensional scapular kinematics in patients with reverse total shoulder arthroplasty during arm motion. *Clin Orthop Surg* 2016;8: 316-24. <https://doi.org/10.4055/cios.2016.8.3.316>
71. Lee SH, Lim KH, Kim JW. Case series of all-arthroscopic treatment for terrible triad of the elbow: indications and clinical outcomes. *Arthroscopy* 2020;36:431-40. <https://doi.org/10.1016/j.arthro.2019.09.014>
72. Lee SH, Kim MB, Lee YH. Simple Osteotomy site repair method for the trans-olecranon approach: Tension Band Wiring with ring Pins. *J Hand Surg Asian Pac Vol* 2021;26:571-9. <https://doi.org/10.1142/S2424835521500545>
73. Leigh WB, Ball CM. Radial head reconstruction versus replacement in the treatment of terrible triad injuries of the elbow. *J Shoulder Elbow Surg* 2012;21:1336-41. <https://doi.org/10.1016/j.jse.2012.03.005>
74. Li T, Li X-L, Hu S-X, Sun W, Wu J. Evaluation of effect and safety of arthroscopic surgery with three different operative approaches in patients with terrible triad of the elbow: a comparative study. *Medicine (Baltim)* 2018;97:e0398. <https://doi.org/10.1097/MD.00000000000010398>
75. Li J, Lu D, Lin W, Li Q, Hu J, Xu D, et al. Prediction and potential preventions for the development of posttraumatic osteoarthritis after the terrible triad injury: a multicenter risk factors study. *BioMed Res Int* 2020;2020:6508781. <https://doi.org/10.1155/2020/6508781>
76. Li D, Song D, Ni J, Tang S, Gao Z, Li P, et al. Single modified posterior approach through the Space of the proximal Radioulnar joint for terrible triad injury: a comparative study. *Orthop Surg* 2022; 14:2159-69. <https://doi.org/10.1111/os.13430>
77. Liu G, Hu J, Ma W, Li M, Xu R, Pan Z. Surgical treatment for terrible triad injury of the elbow with anteromedial coronoid fracture through a combined surgical approach. *J Int Med Res* 2018;46:3053-64. <https://doi.org/10.1177/0300060518771263>
78. Liu G, Ma W, Li M, Feng J, Xu R, Pan Z. Operative treatment of terrible triad of the elbow with a modified pugh standard protocol: retrospective analysis of a prospective cohort. *Medicine (Baltim)* 2018;97:e0523. <https://doi.org/10.1097/MD.00000000000010523>
79. Liu J-Y, Zhang J-Z, Wang Y-M, Tian X, Dong J-M. Difference between posterior monteggia fractures and posterior fracture-dislocation of proximal ulna in adults. *Orthop Surg* 2020;12:1448-55. <https://doi.org/10.1111/os.12784>
80. Longo UG, Franceschi F, Loppini M, Maffulli N, Denaro V. Rating systems for evaluation of the elbow. *Br Med Bull* 2008;87:131-61. <https://doi.org/10.1093/bmb/ldn023>
81. Luengo-Alonso G, García Lamas L, Jiménez Díaz V, Ferrero Recasens J, Porras Moreno MA, Cecilia López D. Trans-olecranon fracture dislocations: how should we treat them? *Injury* 2021;52: S131-6. <https://doi.org/10.1016/j.injury.2021.02.058>
82. Madhar M, Saidi H, Fikry T, Cermak K, Mounghondo F, Schuind F. Dislocation of the elbow with ipsilateral forearm fracture. Six particular cases. *Chir Main* 2013;32:299-304. <https://doi.org/10.1016/j.main.2013.07.004>
83. Maniscalco P, Pizzoli AL, Renzi Brivio L, Caforio M. Hinged external fixation for complex fracture-dislocation of the elbow in elderly people. *Injury* 2014;45:S53-7. <https://doi.org/10.1016/j.injury.2014.10.024>
84. Matar HE, Akimau PI, Stanley D, Ali AA. Surgical treatment of Monteggia variant fracture dislocations of the elbow in adults: surgical technique and clinical outcomes. *Eur J Orthop Surg Traumatol* 2017;27:599-605. <https://doi.org/10.1007/s00590-017-1953-5>
85. Mazhar FN, Ebrahimi H, Jafari D, Mirzaei A. Radial head resection versus prosthetic arthroplasty in terrible triad injury: a retrospective comparative cohort study: a retrospective comparative cohort study. *Bone Jt J* 2018;100-B:1499-505. <https://doi.org/10.1302/0301-620X.100B11.BJJ-2018-0293.R1>
86. Meccariello L, Caiaffa V, Mader K, Prkic A, Eygendaal D, Bisaccia M, et al. Treatment of Unstable elbow injuries with a hinged elbow fixator: subjective and objective results. *Strateg. Trauma Limb Reconstr* 2022;17:68-73. <https://doi.org/10.5005/jp-journals-10080-1553>

87. Najd Mazhar F, Jafari D, Mirzaei A. Evaluation of functional outcome after nonsurgical management of terrible triad injuries of the elbow. *J Shoulder Elbow Surg* 2017;26:1342-7. <https://doi.org/10.1016/j.jse.2017.05.012>
88. Naoki Miyazaki A, Santos Checchia C, Fagotti L, Fregonez M, Doneux Santos P, da Silva LA, et al. Evaluation of the results from surgical treatment of the terrible triad of the elbow. *Rev Bras Ortop* 2014;49:271-8. <https://doi.org/10.1016/j.rboe.2014.03.006>
89. Nestorson J, Josefsson P-O, Adolffson L. A radial head prosthesis appears to be unnecessary in Mason-IV fracture dislocation. *Acta Orthop* 2017;88:315-9. <https://doi.org/10.1080/17453674.2017.1293440>
90. Ngwayi JRM, Tan J, Liang N, Sita EGE, Porter DE. Cross-cultural adaptation and validity of the Chinese version of the Oxford Elbow score. *J Orthop Surg* 2020;15:562. <https://doi.org/10.1186/s13018-020-02100-y>
91. Ohl X, Siboni R. Surgical treatment of terrible triad of the elbow. *Orthop Traumatol Surg Res* 2021;107:102784. <https://doi.org/10.1016/j.otsr.2020.102784>
92. Ostergaard PJ, Tarabochia MA, Hall MJ, Dyer G, Earp BE, Blazar P, et al. What factors are associated with reoperation after operative treatment of terrible triad injuries? *Clin Orthop* 2021;479:119-25. <https://doi.org/10.1097/CORR.0000000000001391>
93. Ozdag Y, Luciani AM, Delma S, Baylor JL, Foster BK, Grandizio LC. Learning curve associated with operative treatment of terrible triad elbow fracture dislocations. *Cureus* 2022;14:e27156. <https://doi.org/10.7759/cureus.27156>
94. Öztürkmen Y, Şükür E, Akman YE, Şenel A, Gürpınar T. Clinical and radiological evaluation of surgical management in olecranon fracture-dislocations. *Musculoskelet Surg* 2020;104:321-8. <https://doi.org/10.1007/s12306-019-00623-0>
95. Papatheodorou LK, Rubright JH, Heim KA, Weiser RW, Sotereanos DG. Terrible triad injuries of the elbow: does the coronoid always need to be fixed? *Clin Orthop* 2014;472:2084-91. <https://doi.org/10.1007/s11999-014-3471-7>
96. Pierrart J, Bégue T, Mansat P. GECC. Terrible triad of the elbow: treatment protocol and outcome in a series of eighteen cases. *Injury* 2015;46:S8-12. [https://doi.org/10.1016/S0020-1383\(15\)70004-5](https://doi.org/10.1016/S0020-1383(15)70004-5)
97. Plaschke HC, Jørgensen A, Thillemann TM, Brorson S, Olsen BS. Validation of the danish version of the Oxford Elbow score. *Dan. Med J* 2013;60:A4714.
98. Pugh DMW, Wild LM, Schemitsch EH, King GJW, McKee MD. Standard surgical protocol to treat elbow dislocations with radial head and coronoid fractures. *JBJS* 2004;86:1122.
99. Reinares F, Rojas N, Calvo A, Aravena C, Rieutord JP, Callejas O, et al. Results of standardized treatment of elbow fracture dislocations as per their injury pattern: a retrospective cohort of 89 patients. *JSES Int* 2021;5:588-96. <https://doi.org/10.1016/j.jseint.2020.12.004>
100. Rodríguez-Quintana D, Comulada DB, Rodríguez-Quintana N, Lopez-Gonzalez F. Radial head ingrowth anatomic implant versus smooth stem monoblock implant in acute terrible triad injury: a prospective comparative study. *J Orthop Trauma* 2017;31:503-9. <https://doi.org/10.1097/BOT.0000000000000885>
101. Sakai K, Shirahama M, Shiba N, Yoshida K, Yoshida S. Primary hinged external fixation of terrible triad injuries and olecranon fracture-dislocations of the elbow. *Kurume Med J* 2017;63:7-14. <https://doi.org/10.2739/kurumemedj.MS6300003>
102. Shen L, Yu X, Zhong B, Ding J. A figure-eight suture loop with Kirschner wires for fixation of anteromedial coronoid process fractures: a case series. *Int J Surg Case Rep* 2017;33:130-4. <https://doi.org/10.1016/j.ijscr.2017.02.016>
103. Sinha S, Sarkar S, Singh A, Saraf SK, Rastogi A, Singh T. Radial head arthroplasty, excision and osteosynthesis in complex elbow fracture-dislocations in young adults: what is preferred? *Indian J Orthop* 2020;54:260-9. <https://doi.org/10.1007/s43465-020-00136-0>
104. Slim K, Nini E, Forestier D, Kwiatkowski F, Panis Y, Chipponi J. Methodological index for non-randomized studies (MINORS): development and validation of a new instrument. *ANZ J Surg* 2003;73:712-6. <https://doi.org/10.1046/j.1445-2197.2003.02748.x>
105. Sørensen AKB, Søjbjerg JO. Treatment of persistent instability after posterior fracture-dislocation of the elbow: restoring stability and mobility by internal fixation and hinged external fixation. *J Shoulder Elbow Surg* 2011;20:1300-9. <https://doi.org/10.1016/j.jse.2011.06.002>
106. Suarez R, Barquet A, Fresco R. Epidemiology and treatment of monteggia lesion in adults: series of 44 cases. *Acta Ortop Bras* 2016;24:48-51. <https://doi.org/10.1590/1413-785220162401152249>
107. Tangtiphaiboonatana J, Agel J, Beingessner D, Hébert-Davies J. Prolonged dislocation and delay to surgery are associated with higher rates of heterotopic ossification in operatively treated terrible triad injuries. *JSES Int* 2020;4:238-41. <https://doi.org/10.1016/j.jseint.2020.02.002>
108. Tanna D. Elbow dislocation with irreparable fracture radial head. *Indian J Orthop* 2013;47:283-7. <https://doi.org/10.4103/0019-5413.111500>
109. Tashjian RZ, Katarincic JA. Complex elbow instability. *J Am Acad Orthop Surg* 2006;14:278-86. <https://doi.org/10.5435/00124635-200605000-00003>
110. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. <https://doi.org/10.1136/bmj.n71>
111. Tille E, Seidel L, Schlüßler A, Beyer F, Kasten P, Bota O, et al. Monteggia fractures: analysis of patient-reported outcome measurements in correlation with ulnar fracture localization. *J Orthop Surg* 2022;17:303. <https://doi.org/10.1186/s13018-022-03195-1>
112. Turner RG, Rispoli D, Lopez-Gonzalez FM, O'Driscoll SW. Partial allograft replacement of the radial head in the management of complex fracture-dislocations of the elbow. *J Shoulder Elbow Surg* 2012;21:396-404. <https://doi.org/10.1016/j.jse.2011.07.003>
113. Wang Y, Huang L, Ma S. Surgical treatment of "terrible triad of the elbow": technique and outcome. *Orthop Surg* 2010;2:141-8. <https://doi.org/10.1111/j.1757-7861.2010.00081.x>
114. Wang W, Liu J, Liu S, Ruan H, Li X, Fan C. Arthrolysis combined with reconstruction for treatment of terrible triad injury with a poor outcome after surgical as well as conservative intervention. *Arch Orthop Trauma Surg* 2014;134:325-31. <https://doi.org/10.1007/s00402-014-1923-x>
115. Watters TS, Garrigues GE, Ring D, Ruch DS. Fixation versus replacement of radial head in terrible triad: is there a difference in elbow stability and prognosis? *Clin Orthop* 2014;472:2128-35. <https://doi.org/10.1007/s11999-013-3331-x>
116. Winter M, Chuinard C, Cikes A, Pelegri C, Bronsard N, de Peretti F. Surgical management of elbow dislocation associated with non-reparable fractures of the radial head. *Chir Main* 2009;28:158-67. <https://doi.org/10.1016/j.main.2009.02.003>
117. Wu H, Liao Q, Zhu Y, Liu H. Surgical reconstruction of comminuted coronoid fracture in terrible triad injury of the elbow. *Eur J Orthop Surg Traumatol* 2012;22:667-71. <https://doi.org/10.1007/s00590-011-0879-6>
118. Wu X, Tang B, Zhu W, Zhou G, Liu F, Qiu B, et al. Evaluation of a combined posterior lateral and anteromedial approach in the treatment of terrible triad of the elbow: a retrospective study. *Medicine (Baltim)* 2017;96:e6819. <https://doi.org/10.1097/MD.00000000000006819>
119. Wyrick JD, Dailey SK, Gunzenhaeuser JM, Casstevens EC. Management of complex elbow dislocations: a mechanistic approach. *J Am Acad Orthop Surg* 2015;23:297-306. <https://doi.org/10.5435/JAAOS-D-14-00023>
120. Yan M, Ni J, Song D, Ding M, Liu T, Huang J. Radial head replacement or repair for the terrible triad of the elbow: which procedure is better? *ANZ. J Surg* 2015;85:644-8. <https://doi.org/10.1111/ans.13060>
121. Zaidenberg EE, Abrego MO, Donndorff AG, Boretto JG, De Carli P, Gallucci GL. Treatment of terrible triad injuries at a mean follow-up

- of nine years. *Shoulder Elb* 2019;11:450-8. <https://doi.org/10.1177/1758573218809375>
122. Zarezadeh A, Mamelson K, Thomas WC, Schoch BS, Wright TW, King JJ. Outcomes of distal humerus fractures: what are we measuring? *Orthop Traumatol Surg Res* 2018;104:1253-8. <https://doi.org/10.1016/j.otsr.2018.08.017>
123. Zha Y-J, Xiao D, Hua K-H, Sun W-T, Gong M-Q, Li T, et al. Lateral minimal approach to the terrible triad of the elbow: a treatment protocol in Beijing Jishuitan Hospital. *Ann Transl Med* 2021;9:1232. <https://doi.org/10.21037/atm-21-2542>
124. Zhang C, Zhong B, Luo C-F. Treatment strategy of terrible triad of the elbow: experience in Shanghai 6th People's Hospital. *Injury* 2014;45:942-8. <https://doi.org/10.1016/j.injury.2013.12.012>
125. Zhang J, Tan M, Kwek EBK. Outcomes of coronoid-first repair in terrible triad injuries of the elbow. *Arch Orthop Trauma Surg* 2017; 137:1239-45. <https://doi.org/10.1007/s00402-017-2733-8>
126. Zhou C, Lin J, Xu J, Lin R, Chen K, Sun S, et al. Does timing of surgery Affect treatment of the terrible triad of the elbow? *Med Sci Monit* 2018;24:4745-52. <https://doi.org/10.12659/MSM.907146>